Muhammad Faisal

Full Stack MERN Developer · AI Integration Engineer

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PROFESSIONAL SUMMARY

Final-year Software Engineering student (CGPA 3.5/4.0) with production-shipped experience across the full MERN stack and Python AI services. Built and deployed three real-world SaaS products — an AI background-removal tool, a financial intelligence platform, and a habit/productivity app — all as sole developer from architecture through deployment. Comfortable bridging React/Next.js frontends, Node.js/Express APIs, MongoDB data models, and Python FastAPI microservices. Targeting a junior MERN or AI-integrated full-stack role where I can ship fast and own meaningful product surface.

CORE SKILLS

Frontend
React.js · Next.js 15
JavaScript (ES2023) · TypeScript
Tailwind CSS · HTML5 · CSS3
PWA · Responsive Design

Backend & Database
Node.js · Express.js · REST APIs
MongoDB · Mongoose · SQL
JWT Auth · Clerk · Cloudinary
Multer · File Handling

AI & DevOps
Python · FastAPI · LangChain
rembg · ONNX Runtime · U2Net
Git · GitHub · Vercel · Postman
DSA · OOP · DBMS · Agile

PROJECTS

CutOut Studio — AI Background Removal SaaS for E-Commerce

Next.js 15 · React 19 · Node.js · Express · FastAPI · Python · rembg (U2Net) · ONNX · Tailwind CSS · MulterLive · GitHub

A zero-cost-per-image background removal platform with two modes: single-image processing with a before/after comparison slider, and bulk processing of up to 50 images simultaneously with ZIP download. Runs a local Python/FastAPI AI service (rembg + U2Net), so removal is free at any volume.

Architecture:
Three-tier system: Browser → Next.js 15 API routes → Node.js/Express proxy → Python FastAPI/rembg microservice. The AI service is independently deployable for horizontal scaling under heavy load.

My Role:
Sole developer — UI components, AI integration, API architecture, deployment configuration.

Impact:

- **$0 per-image cost** vs remove.bg's $0.20/image. A seller processing 500 images/month saves $100/month.
- Bulk mode cuts processing from hours to minutes — 50 images queued, processed, and downloaded as a ZIP.
- Modular microservice architecture allows independent AI scaling without redeploying the Next.js frontend.

FinancePulse — AI-Powered Financial Intelligence Platform for SMEs

React.js · Node.js · Express · MongoDB · Chart.js · JWT Auth · LangChain · OpenAI API

A financial intelligence dashboard that lets small and medium businesses track income, expenses, and cash flow in real time, with an AI layer that surfaces anomalies, forecasts trends, and answers natural-language queries about their own data.

Architecture:
MERN stack with a LangChain agent connected to MongoDB aggregation pipelines. Users can ask 'What was my highest-expense month?' and receive chart + prose answers generated against live data.

My Role:
Sole developer — schema design, REST API, LangChain integration, charting dashboard, JWT auth flow.

Impact:

- Natural-language query engine eliminates manual report generation — answers in under 2 seconds against live MongoDB data.

- **Anomaly detection** flags expense spikes automatically, surfacing patterns a business owner would otherwise miss in spreadsheets.
- Modular chart components (Chart.js) render income vs. expense breakdowns, trend lines, and category distributions from a single API call.

TaskFlow — Discipline-Oriented Productivity & Habit Tracking Platform

React.js · Node.js · Express · MongoDB · PWA · Web Notifications API · Web Speech API Live · GitHub

A full-stack productivity system built around tasks, recurring habits, time-block scheduling, and smart search — designed for users who want accountability over convenience. Ships as a Progressive Web App for offline-capable, mobile-first use.

Architecture:

MERN stack with a service-worker PWA layer. Habit recurrence is handled server-side via a scheduled job; analytics aggregations run on MongoDB aggregation pipelines and are cached client-side.

My Role:

Sole developer — UI/UX design, REST API, database schema, PWA configuration, analytics dashboards, voice input integration.

Impact:

- Analytics dashboard with streak graphs and completion-rate charts gives users a quantified view of consistency over time.
- **Voice input** (Web Speech API) and browser push notifications create a hands-free capture loop that reduces friction for on-the-go task entry.
- PWA install delivers app-like experience with offline support — no App Store dependency.

EDUCATION

Islamia University of Bahawalpur (IUB) Expected 2027

Bachelor of Software Engineering · **CGPA 3.5 / 4.0**

Relevant coursework: Data Structures & Algorithms · OOP · Database Management Systems · Operating Systems · Computer Networks · Software Engineering

CERTIFICATIONS

Agile Project Management Certification HP (Hewlett-Packard)